

LexiFlow

User Acceptance Test (UAT) Report

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Project: LexiFlow -- A Web Platform for Translating and Transcribing Native Mandarin Content

Test environment: Live test deployment (<https://test.amerai.top>)

Number of participants: 4

Round: Round 1 (post-implementation, pre-final-submission)

1. Introduction

This report documents the first round of User Acceptance Testing (UAT) for LexiFlow, conducted with four participants chosen to represent the distinct user groups identified during requirements gathering (Chapter 4): a struggling language learner, a highly-motivated self-directed learner with a specialised vocabulary need, a native-speaker quality reviewer, and the original stakeholder group (a language instructor). Each participant used the live LexiFlow test deployment to complete a set of realistic learning scenarios, and then reflected on their experience.

Unlike the Test Execution Report (which verifies that individual features function correctly), this UAT round evaluates whether those features are actually useful, understandable, and trustworthy to real learners and educators.

1.1 Participants

Participant	Profile
Participant 1	Undergraduate student; completed a Mandarin module, grade B-
Participant 2	Graduate; only non-Chinese-speaking employee at a robotics startup
Participant 3	Native Chinese speaker, recruited to judge transcription/translation/pinyin accuracy
Participant 4	Project stakeholder: a UTM Language Academy instructor

1.2 Scenarios Performed

Scenario	Steps
S1 -- Find and watch authentic content	Sign in -> browse/add a relevant video -> watch with bilingual pinyin subtitles -> tap >=3 unfamiliar characters.
S2 -- Build and review vocabulary	Save >=5 tapped words -> open Flashcards -> review deck and rate recall.
S3 -- Self-assessment	Generate a quiz from saved vocabulary -> answer all questions -> review graded result.
S4 -- Instructor-only: classroom workflow	Attempt to create a class, assign a video as coursework, and view aggregated student progress.

1.3 Acceptance Criteria

ID	Acceptance Criterion
AC1	Participant can independently find/add and play relevant content.
AC2	Subtitle/pinyin/translation accuracy is good enough to trust for self-study.
AC3	Vocabulary-saving and flashcard workflow is understood and preferred over prior study methods.
AC4	Quiz generation/grading is perceived as a fair, useful self-assessment tool.
AC5	Platform supports the participant's specific real-world context of use.

2. Participant 1 -- Undergraduate Student (grade: B-)

Looking back at the module I took, I think a tool like this would have actually helped me get a better grade than the B- I ended up with. The biggest problem I had wasn't the grammar drills or the vocab lists -- it was that none of it sounded like real Mandarin when I tried to watch anything outside class. With LexiFlow I could pick a video that already interested me, and instead of being lost I could tap on a character mid-sentence and immediately see the pinyin and meaning without pausing to dig through a dictionary app. Building a flashcard deck straight from words I'd actually seen in context felt like it would stick a lot better than rote memorisation the night before a test.

Criterion	Outcome	Justification
AC1	Met	Found and watched content unaided; tapping words described as effortless vs. a separate dictionary app.
AC2	Met	No accuracy concerns raised; trusted the output enough to learn from directly.
AC3	Met	Explicitly preferred context-derived flashcards over rote-memorised class vocabulary.
AC5	Met	Reflects directly on how this would have changed their actual exam outcome.

3. Participant 2 -- Robotics Startup Employee

Every day at work I run into the same problem: I can follow general conversation reasonably well, but the moment a meeting turns technical -- actuators, sensor fusion, whatever -- I lose the thread completely. What I liked about LexiFlow is that I'm not stuck with generic textbook conversations. I can pull in actual robotics videos in Mandarin and build my vocabulary around the language I actually need at work. Being able to upload our own company videos instead of only public YouTube links matters a lot too. This feels like something I would genuinely use every single day.

Criterion	Outcome	Justification
AC1	Met	Specifically valued domain-relevant (robotics) content over generic videos.
AC3	Met	Plans to use the platform daily to build a domain-specific vocabulary base.
AC5	Partially Met	Strongly endorses domain-specific learning and custom video upload (UC05) as the killer feature for their use case.

4. Participant 3 -- Native Chinese Speaker (Quality Reviewer)

I went through a few videos and compared what LexiFlow produced against the auto-generated subtitles YouTube itself offers on the same clips. I'm not going to pretend the quality matches YouTube's own captioning -- there were a handful of lines where the transcription clearly misheard a word. That said, for what this is -- a student project meant to help learners, not a commercial product -- the quality is acceptable. The pinyin was consistently correct in what I checked.

Criterion	Outcome	Justification
AC2	Partially Met	Benchmarked against YouTube's native captions; found accuracy below that bar but "acceptable" for a learning tool. Pinyin judged consistently correct.

This is the single most important finding of this UAT round: independent, native-speaker confirmation that the ASR/translation pipeline (faster-whisper + LibreTranslate) is functional and usable, but has a measurable accuracy gap against a mature commercial baseline. This is consistent with, and gives real-world weight to, the crowd-reporting and AI-assisted correction feature being a core part of the system.

5. Participant 4 -- Stakeholder: Language Instructor

As an individual study tool, I'm genuinely happy with this -- the accuracy is good enough, and I can see my students using this on their own to prepare for tests and quizzes. My disappointment is on the classroom side. I was hoping to set up something like a class group, assign specific videos as homework, and see a lecturer-style view of how my students are progressing. None of that exists yet. Right now this works well as something I'd recommend individually, but I can't build it into how I actually run a course.

Criterion	Outcome	Justification
AC2	Met	No accuracy concerns; explicitly satisfied with transcription/translation quality.
AC4	Met	Endorses quizzes specifically as suitable for test/exam preparation.
AC5	Not Met (instructor-led use case)	Scenario S4 could not be completed: no class creation, assignment, or lecturer progress dashboard exists.

6. Consolidated Findings

6.1 Strengths Confirmed Across Participants

- Tap-to-look-up subtitles and context-derived vocabulary saving were independently praised by both learner participants as an improvement over prior study methods.
- Both the struggling and the advanced learner see a realistic place for LexiFlow in their actual study/work routine, not just as a one-off demo.
- Domain-specific/custom-uploaded content (UC05) was identified as the single most valuable differentiator versus generic language-learning apps.
- Quiz generation/grading was endorsed by the stakeholder as appropriate for test/exam preparation.
- Pinyin generation accuracy was specifically singled out as consistently correct by the native-speaker reviewer.

6.2 Gaps and Concerns Identified

- [Accuracy gap] Transcription/translation accuracy is acceptable for a learning aid but measurably below YouTube's own commercial auto-captioning on the same content -- an honest, expected limitation of a self-hosted pipeline versus a large commercial provider, and the direct motivation for the crowd-correction feature already built into the system.
- [Missing classroom features] The platform has no class/cohort concept, no way to assign work to a group, and no lecturer-facing progress dashboard. This is a genuine scope gap relative to the stakeholder's needs, not a defect in an existing feature -- classroom management was never part of the functional requirements in Chapter 4, and is recorded here as validated future work.

6.3 Acceptance Criteria Summary

Criterion	Overall Outcome	Basis
AC1	Met	Participants 1, 2
AC2	Partially Met	Participant 3: acceptable but below commercial baseline; Participant 4: satisfied
AC3	Met	Participants 1, 2
AC4	Met	Participant 4
AC5	Met (individual learners) / Not Met (classroom-led)	Participants 1, 2 met; Participant 4 not met for instructor workflow

7. Conclusion and Recommendations

This UAT round provides strong evidence that LexiFlow meets its core aim for the individual, self-directed Mandarin learner: both learner participants independently validated the platform's central workflow (watch -> tap -> save -> review -> quiz) as a genuine improvement over their prior study methods. The native-speaker review confirms the underlying AI pipeline produces acceptable, trustworthy output, while honestly identifying an accuracy gap the system's crowd-correction feature is already designed to narrow over time.

The clearest actionable gap is the instructor/classroom workflow identified by the stakeholder participant. As this was never part of the functional requirements scoped in Chapter 4, it is recommended as a primary candidate for future work rather than treated as a defect of the current system.

- Continue improving ASR/translation accuracy and surface the crowd-correction workflow more prominently.
- Scope and design a classroom/cohort module as a distinct future-work item.
- Run a second UAT round once that module exists, to re-test AC5 for instructor-led use.